New Uzbekistan University

Statement of Ethics: I agree to complete this exam without unauthorized assistance.

FULL NAME: SIGNATURE:

ID NUMBER: EXAM ROOM: M A T H GROUP:

Final Exam

COURSE NAME: Microeconomics 3 COURSE NUMBER: ECON630

EXAMINATION DURATION: 120 MINUTES EXAM VERSION A

ADDITIONAL MATERIALS ALLOWED TO USE: Solve all 4 problems. Calculators or other materials are not allowed.

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST! VIOLATION OF THE RULES CAN LEAD TO A LOSS OF POINTS.

- Read and strictly follow extra instructions at the beginning of each Section.
- Put your signature and write your ID on even pages.
- Follow the University Examination Rules.

FOR INSTRUCTOR USE ONLY (DO NOT WRITE ANYTHING):

Question:	1	2	3
Points:	25	25	25
Score:			

Question:	4		Total
Points:	25		100
Score:			

1 Problems MOCK EXAM

Solve problems. Show your work.

1. (25 points) Robinson and Friday have identical utility function $u(x, y) = \min(x, y)$. Endowment of Robinson is (a, b), $a \neq b$, endowment of Friday is (b, a).

   - Explain why there is no equilibrium if Robinson is alone on the island.
   - Is there an equilibrium if both Robinson and Friday are on the island? What are the prices?

2. (25 points) Assume there are three consumers in Stackelberg's model, with the leader and two followers. Followers choose their outputs simultaneuosly after the leader chose their output. Assume the linear demand $P = A - BQ$ and constant marginal costs c. Solve for the equilibrium output.

3. (25 points) A consumer has wealth of 25. There is a 50% probability he'll lose 9. Calculate the expected utility, the utility of the average outcome, and the most the insurance company can charge this consumer to completely insure away the risk.

4. (25 points) There are three consumers with identical endowments of $\omega = (0, 3)$ and utility functions

$$u_1(x_1, y_1|x_2, x_3) = \ln(x_1 + \alpha x_2 - \alpha x_3) + y_1,$$
$$u_2(x_2, y_2|x_1, x_3) = \ln(x_2 + \alpha x_3 - \alpha x_1) + y_2,$$
$$u_3(x_3, y_3|x_1, x_2) = \ln(x_3 + \alpha x_1 - \alpha x_2) + y_3.$$

x can be produced out of y with a technology $x - y \leq 0$.

- What relative price of x to y can we expect in the equilibrium?
- What is the market equilibrium?
- Is the market equilibrium Pareto efficient?